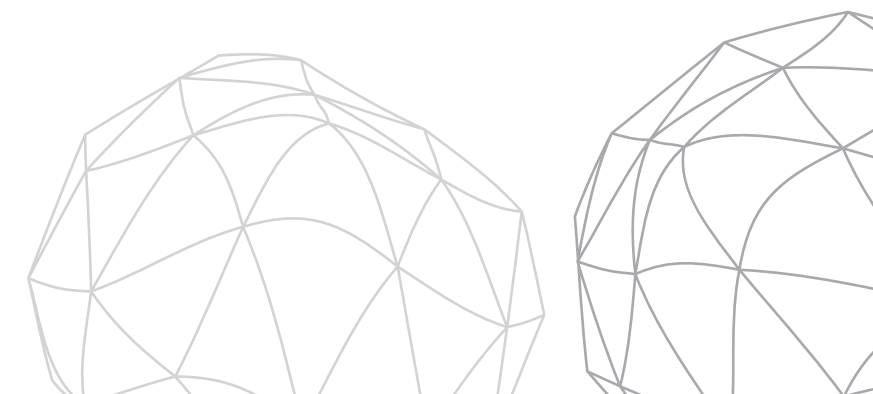

Program
Računarstva

O računanju, računarima i računarstvu

Stefan Nožinić



**Potreba čovečanstva da nešto
izračuna**



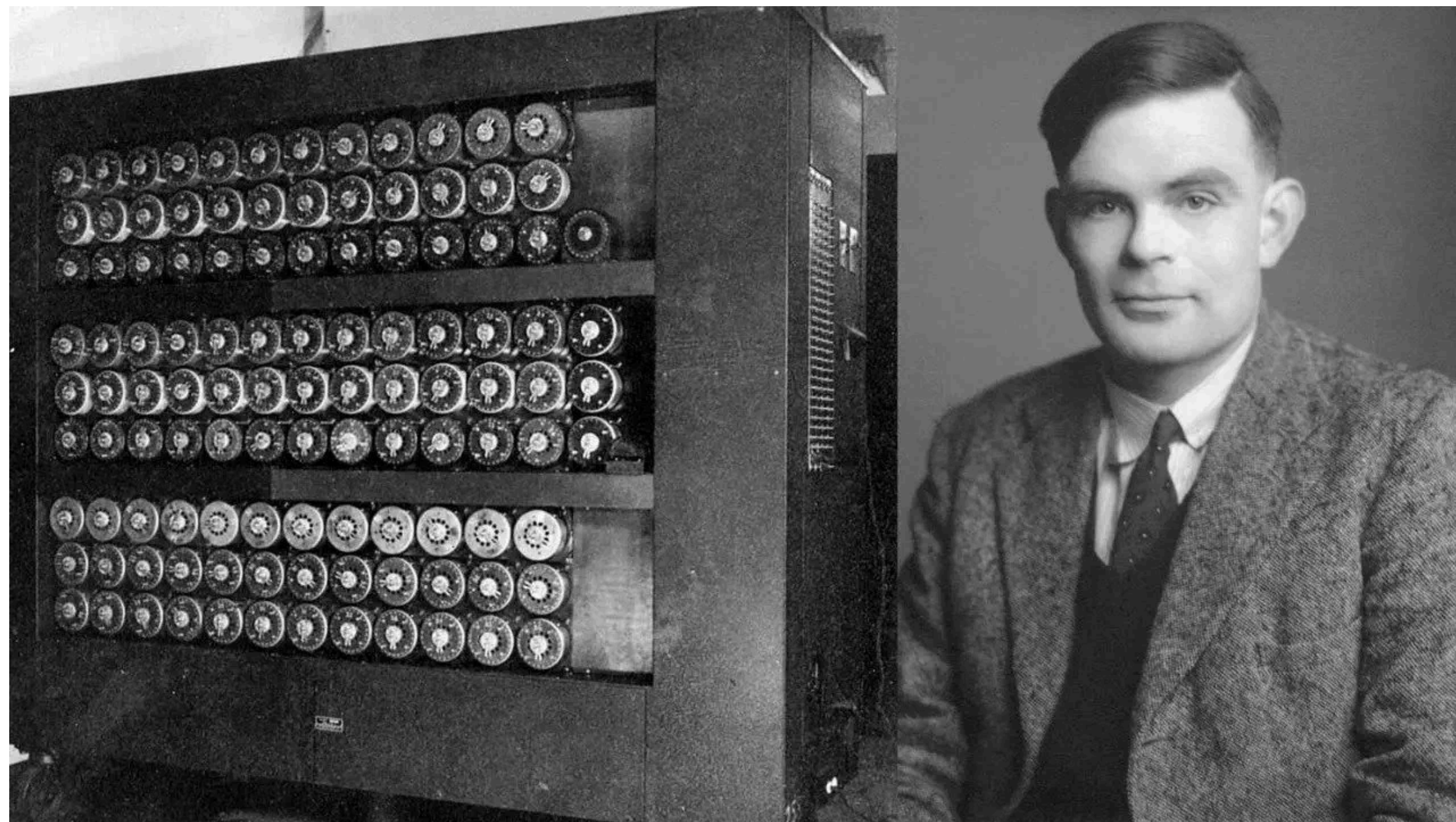
```
graph LR; A[Instanca] --> B[Problem]; B --> C[Rešenje];
```

Instanca

Problem

Rešenje

Alan Turing



A-machine

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHIEDUNGSPROBLEM

By A. M. TURING.

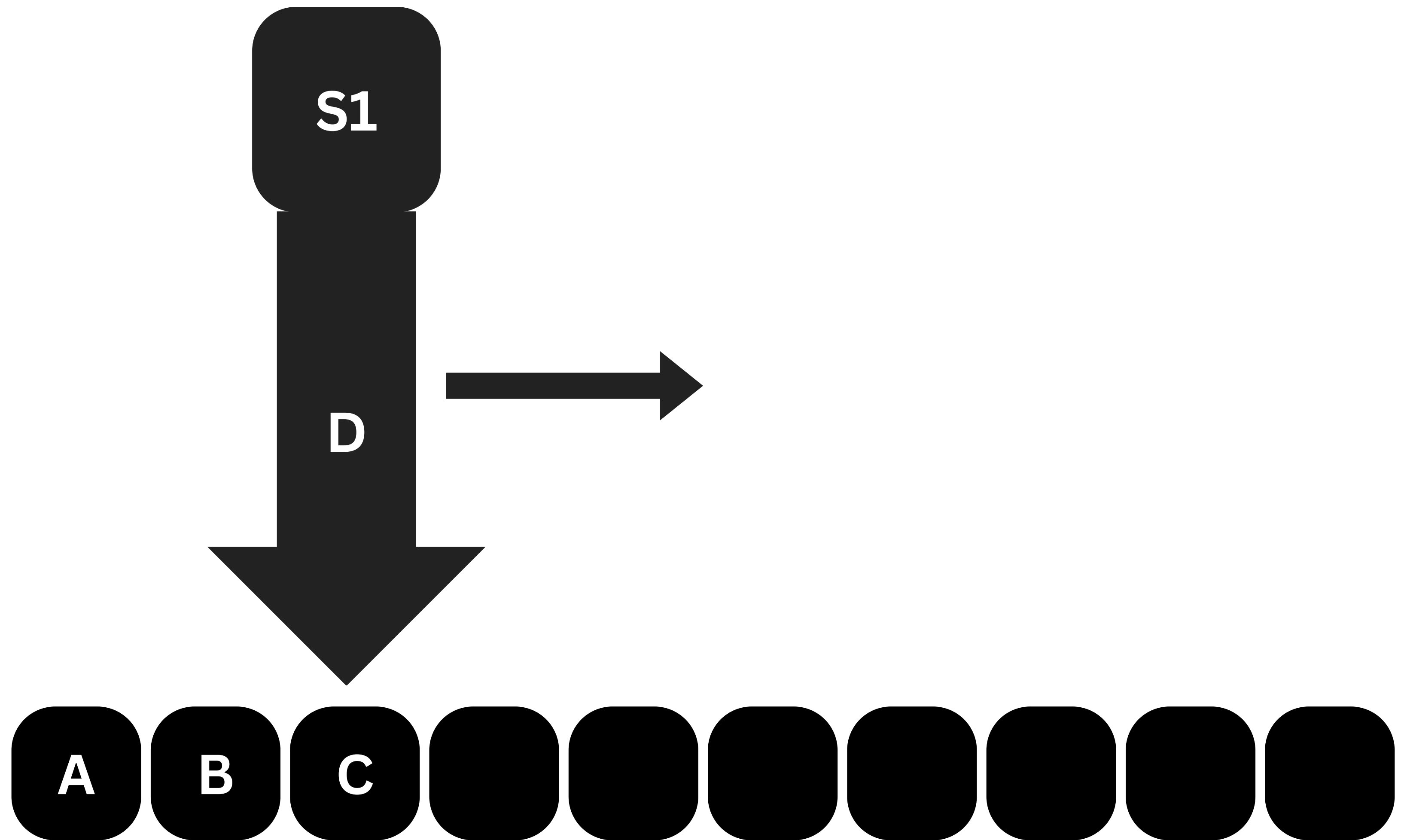
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The “computable” numbers may be described briefly as the real numbers whose expressions as a decimal are calculable by finite means. Although the subject of this paper is ostensibly the computable *numbers*, it is almost equally easy to define and investigate computable functions of an integral variable or a real or computable variable, computable predicates, and so forth. The fundamental problems involved are, however, the same in each case, and I have chosen the computable numbers for explicit treatment as involving the least cumbrous technique. I hope shortly to give an account of the relations of the computable numbers, functions, and so forth to one another. This will include a development of the theory of functions of a real variable expressed in terms of computable numbers. According to my definition, a number is computable if its decimal can be written down by a machine.

In §§ 9, 10 I give some arguments with the intention of showing that the computable numbers include all numbers which could naturally be regarded as computable. In particular, I show that certain large classes of numbers are computable. They include, for instance, the real parts of all algebraic numbers, the real parts of the zeros of the Bessel functions, the numbers π , e , etc. The computable numbers do not, however, include all definable numbers, and an example is given of a definable number which is not computable.

Although the class of computable numbers is so great, and in many ways similar to the class of real numbers, it is nevertheless enumerable. In § 8 I examine certain arguments which would seem to prove the contrary. By the correct application of one of these arguments, conclusions are reached which are superficially similar to those of Gödel†. These results

† Gödel, “Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme, I”, *Monatshefte Math. Phys.*, 38 (1931), 173–198.



Formalno

$$M = (Q, \Gamma, b, \Sigma, \delta, F)$$

$$\delta : (Q \setminus F) \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

Šta TM mašina sve može?

- Da prepozna palindrome
- Da sabere brojeve
-
- Da simulira TM

Algoritam i enkodiranje

- **Algoritam je program koji izračunava, odnosno rešava neki problem**
- **Ulaz mu je instanca problema**
- **Izlaz je rešenje problema**
- **Formalno, pravimo Tjuringovu mašinu ili ekvivalentno (malo više o tome kasnije)**
- **Enkodiranje je proces gde instancu problema svodimo na simbole na traci TM**

Ali... moj računar nije to!

- **Današnji računari se oslanjaju na drugi model računanja - RAM - Random Access Machine**
- **U RAM modelu, pristup bilo kom elementu unutar memorije je brz, za razliku od Tjuringove mašine gde je pristup sekvencijalan**

Tjuring kompletna mašina

- Bilo kakva mašina, odnosno model, koji može da simulira Tjuringovu mašinu
- Vaš računar
- Game of Life
- Kompajliranje C++-a 😊
- Čelija! 😊

Tjuring ekvivalentna mašina

- **Bilo kakva mašina, odnosno model, koji je Tjuring-kompletna i koji se može simulirati Tjuringovom mašinom**
- **Dakle, sve ono što nema veću računarsku moć od Tjuringove mašine**
- **Npr: RAM, vaš računar, kvantni računar, ...**

A šta TM ne može?

- Tjuringova mašina može da simulira drugu Tjuringovu mašinu, ali ne može da odluči da li druga Tjuringova mašina će da stane sa radom i dođe u finalno stanje
- Ovaj problem se zove Halting problem
- Teorijski je nemoguće rešiti ovaj problem sa trenutnim modelima računanja

Dva glavna pitanja

- **Koliko je algoritam složen?**
- **Da li je algoritam tačan?**

Problemi odlučivanja

- **Rešenje je odgovor DA ili NE**
- **Primeri:**
 - **Da li je dati string palindrom**
 - **Da li je dati broj paran**
 - **Da li je broj deljiv sa 3**

SAT

.

Klase problema

- Klasa P
- Klasa NP
- $P = NP$ problem
- NP-hard
- NP-complete
- EXP

3CNF SAT

- Oblika: $C_1 \text{ AND } C_2 \text{ AND } \dots C_k$
- Svako C_i sadrži disjunkciju 3 literala
- Literal je varijabla ili njena negacija
- Isto NP-complete

Problem klike

- Dat je graf G
- Klika je kompletan podgraf G
- Problem: Da li G ima kliku?
- Svodimo na 3CNF SAT

**Distribuirani
algoritmi**

**Slučajni
algoritmi**

**Paralelni
algoritmi**

**Aproks.
algoritmi**

Strukture podataka

Programski jezici

Arhitektura računara

Teorija izračunavanja

Q&A